

1. **Which header explains why Isabella received the email without being rejected by the email platform?**
2. **What technique did the attacker use to make the message seem legitimate?**
3. **Which MITRE technique and sub-technique ID best fit this sender address trick?**
4. **What is the file extension of the attached file?**
5. **What is the MD5 hash of the .HTML file?**
6. **What is the landing page of the phishing attack?**
7. **Which MITRE technique ID was used inside the attached file?**
8. **What is the hidden message the attacker left in the file?**
9. **Which line in the attached file is responsible for decoding the URL redirect?**
10. **What is the first URL in the redirect chain?**
11. **What is the Threat Actor associated with this malicious file and/or URL?**
12. **What is the main target of this Threat Actor according to MITRE?**

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1. Which header explains why Isabella received the email without being rejected by the email platform?

The email had SPF, DKIM, and DMARC all set to none. DMARC is the most important for us because it tells the receiving server what to do with emails that fail authentication. With p=none, it only monitors emails and doesn't request blocking them.

Key	Value
delivered-to	isabella@kingford.ac.uk
received	by 2002:a05:7022:e2a:b0:119:5273:6922 with SMTP id dx42csp217860dlb; Tue, 4 Nov 2025 06:46:55 -0800 (PST) from mail-sor-f41.kinglord.ac.uk (mail-sor-f41.kinglord.ac.uk. [207.84.120.31]) by mx.kinglord.ac.uk with SMTPS id a640c23a62f3a-b715aee65casor170440966b.13.2025.11.04.06.46.54 for <isabella@kingford.ac.uk> (Google Transport Security); Tue, 04 Nov 2025 06:46:55 -0800 (PST)
x-received	by 2002:a17:907:7241:b0:b70:b13c:3622 with SMTP id a640c23a62f3a-b70b13c60c3mr1024470066b.4.1762267614032; Tue, 04 Nov 2025 06:46:54 -0800 (PST)
arc-seal	i=1; a=rsa-sha256; t=1762267615; cv=none; d=kinglord.ac.uk; s=arc-20240605; b=BHopKeGbF2EaIKddox+LMO08kRWGczIGa1tLrrYAhKwk7WbsnsGS0cQKRJYMqssw6D du5B4U0n5zRiT25iQhXjRASjK6+yjSUdI4Y3yCAsUZ+xJzH+xBP9yYuDxIJlYwc 1tC672IOx3RYNBcln5lU9GCm2pIa7aMBINFdijpg8wPFifTYIDjq9L342ChHkeZ98IwP FL1LyPIQww9TQhr/4u+WgZC4DA932DYHXloX60UM00jPoXxyEIOCO5HO4ARS1XqMJf/B ErkI9HI1tMICYSId9mVvLiVh71ScInBVNTgpyNEhWxffPCTUTmvU7jUoHy/
arc-message-signature	i=1; a=rsa-sha256; c=relaxed/relaxed; d=kinglord.ac.uk; s=arc-20240605; h=to:subject:message-id:date:from:mime-version:dkim-signature; bh=eHJylYXSdwWABXLpJYYVpVmG3UlqRqj1JS3MwUoLrF0=; fh=X/71MsKVOFuVb6a2euNxQ7KQPFXo6IZb b=lsQ89IJN3R2RAnjxPHdI5ykuyq0O6H3lZEgLRGSuNarNCOuDBrbqxdrLq25GLzI8nA I+IXCuUgnv+fskGzCvAZSwtMEw2VAv+OjxMZOjv08hxIC/DJrJFsAP68IpcnKYWKg0Lo 2c1T4ytOuMpIRhaLI3fLsWp+BQ3QFENm2N8tEBGdTutPYNsEp/9Nh86KrCo PfP2zzIKpqkpjm6nIZJ9mt4V00DmI4qIGV8OPxfvXQfem3o3CAPC55Xe7nM/I9e8kvNm /ILMrvpZI7Doht8g/y48T35gDysb02NyIiBS4wfOlVPXzu4+Q26laKJradBnyEW0NLZ+ uCiw==; dara=kinglord.ac.uk
arc-authentication-results	i=1; mx.kinglord.ac.uk; dkim=none (no DKIM signature) header.i=@kinglord.ac.uk header.s=20230601 header.b=mMtxjlwD; spf=none (kinglord.ac.uk: no SPF record) smtp.mailfrom=library@kinglord.ac.uk; dmarc=none (kinglord.ac.uk: no DMARC recor dara=neutral header.i=@kingford.ac.uk

Answer: DMARC=none

2. What technique did the attacker use to make the message seem legitimate?

The email was sent from a suspicious domain impersonating our legitimate domain, which is consistent with a typosquatting attack

Answer: Typosquatting

3. Which MITRE technique and sub-technique ID best fit this sender address trick?

We should search the MITRE ATT&CK website for a technique related to typosquatting domains

Acquire Infrastructure: Domains

Other sub-techniques of Acquire Infrastructure (8)

Adversaries may acquire domains that can be used during targeting. Domain names are the human readable names used to represent one or more IP addresses. They can be purchased or, in some cases, acquired for free.

Adversaries may use acquired domains for a variety of purposes, including for Phishing, Drive-by Compromise, and Command and Control.[1] Adversaries may choose domains that are similar to legitimate domains, including through use of homoglyphs or use of a different top-level domain (TLD).[2][3] Typosquatting may be used to aid in delivery of payloads via Drive-by Compromise. Adversaries may also use internationalized domain names (IDNs) and different character sets (e.g. Cyrillic, Greek, etc.) to execute "IDN homograph attacks," creating visually similar lookalike domains used to deliver malware to victim machines.[4][5][6][7][8]

ID: T1583.001

Sub-technique of: T1583

ⓘ Tactic: Resource Development

ⓘ Platforms: PRE

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Answer: T1583.001

4. **What is the file extension of the attached file?**

We'll continue analyzing the email provided in the task to identify the attachment

Data

Key	Value
1	library-invoice.pdf.html

Answer: .HTML

5. **What is the MD5 hash of the .HTML file?**

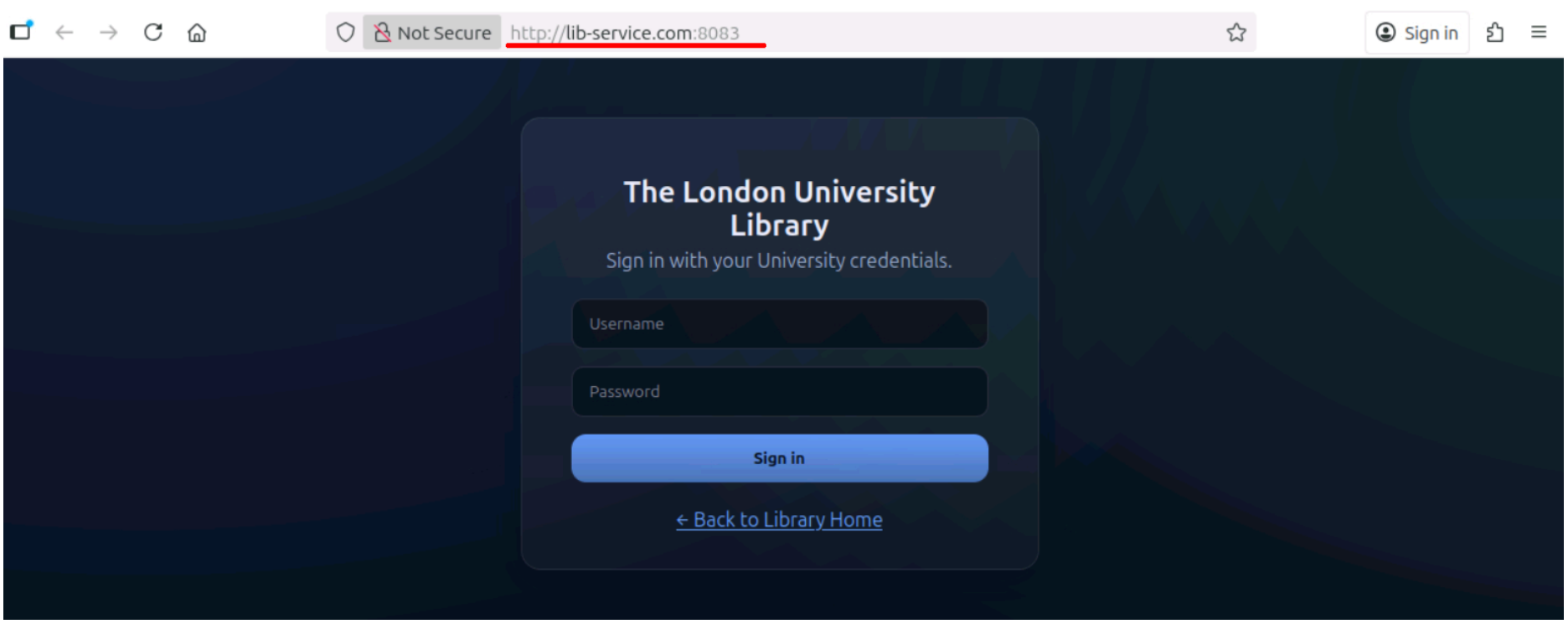
We're checking the MD5 hash of the file attached to the email

```
ubuntu@tryhackme:~/Desktop$ md5sum library-invoice.pdf.html
442f2965cb6e9147da7908bb4eb73a72  library-invoice.pdf.html
```

Answer: 442f2965cb6e9147da7908bb4eb73a72

6. **What is the landing page of the phishing attack?**

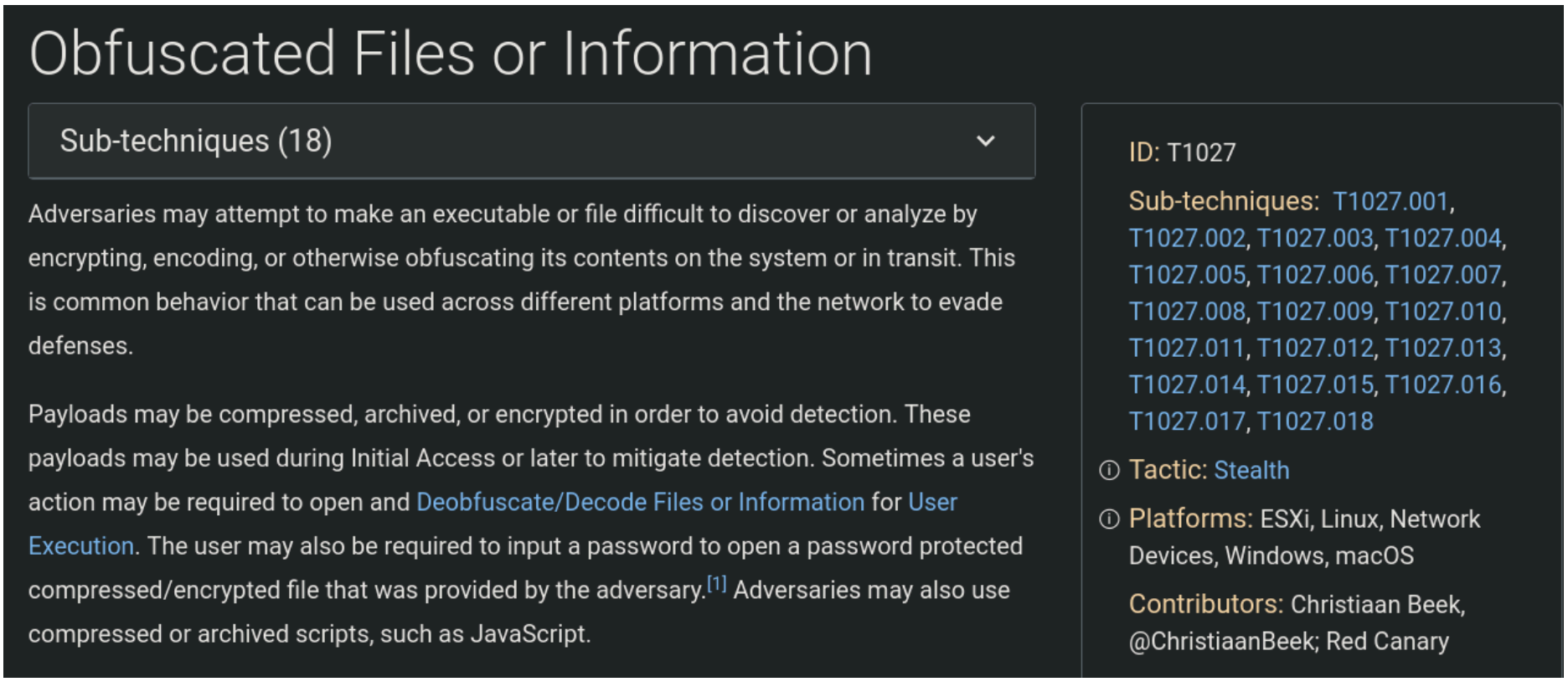
I'm checking which page the suspicious file redirects us to



Answer: http://lib-service.com:8083

7. **Which MITRE technique ID was used inside the attached file?**

The file contains a JavaScript script with Unicode-obfuscated values, which is consistent with MITRE ATT&CK T1027 (Obfuscated Files or Information)



Answer: T1027

8. **What is the hidden message the attacker left in the file?**

After deobfuscating the script, we found the previously obfuscated message and a redirect URL

main.js

Run ▶

```
const xanthium = [
  "\u0032\u0038\u0030\u0038\u003a",
  "\u006d\u006f\u0063\u002e",
  "\u0032\u0033\u0034\u0032\u0033\u0065\u0077\u0063\u0033\u0031\u002d",
  "\u0075\u006c\u0074\u0079\u0072\u0061\u0072\u0062\u0069\u0456\u006c",
  "\u002f\u002f\u003a\u0070\u0074\u0074\u0068"
];

const egassem = [
  "\u005e\u005e\u0020\u0073\u0065\u0069\u0072\u0061\u0072",
  "\u0062\u0069\u006c\u0020\u006d\u006f\u0072\u0066\u0020",
  "\u0073\u006b\u006f\u006f\u0062\u0020\u0068\u0073",
  "\u0069\u0068\u0070\u0020\u006f\u0074\u0020",
  "\u0065\u0076\u006f\u006c\u0020\u0049"
];

const src = xanthium.join("").split("").reverse().join("");
const src2 = egassem.join("").split("").reverse().join("");

console.log("1:", src);
console.log("2:", src2);
```

Output

```
1: http://liibrarytlu-13cwe32432.com:8082
2: I love to phish books from libraries ^^
```

Answer: I love to phish books from libraries ^^

9. Which line in the attached file is responsible for decoding the URL redirect?

To answer this question, we need to analyze the file's code

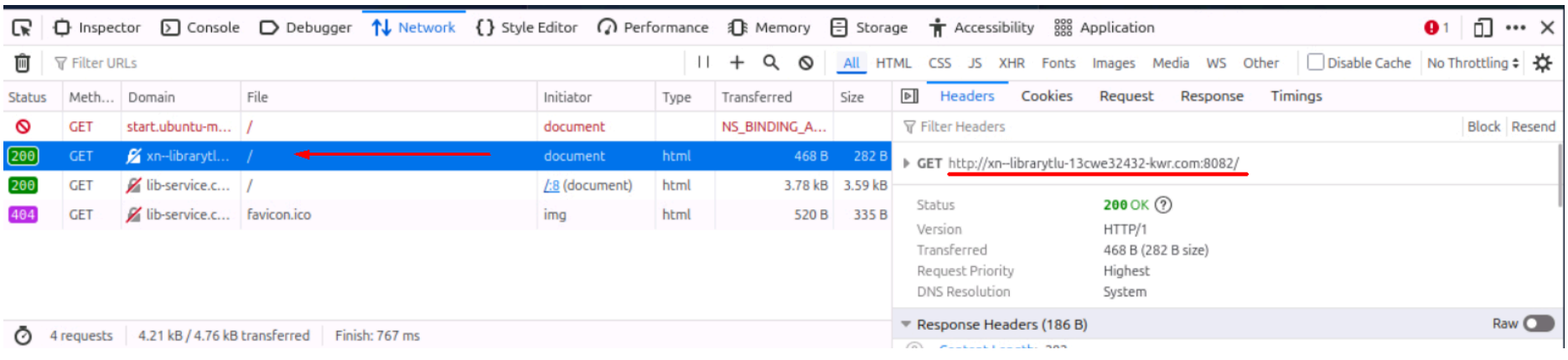
```
var reversed = xanthium.join("");
var src = reversed.split("").reverse().join("");

var reversed = egassem.join("");
var mes = reversed.split("").reverse().join("");
```

Answer: var src = reversed.split("").reverse().join("");

10. What is the first URL in the redirect chain?

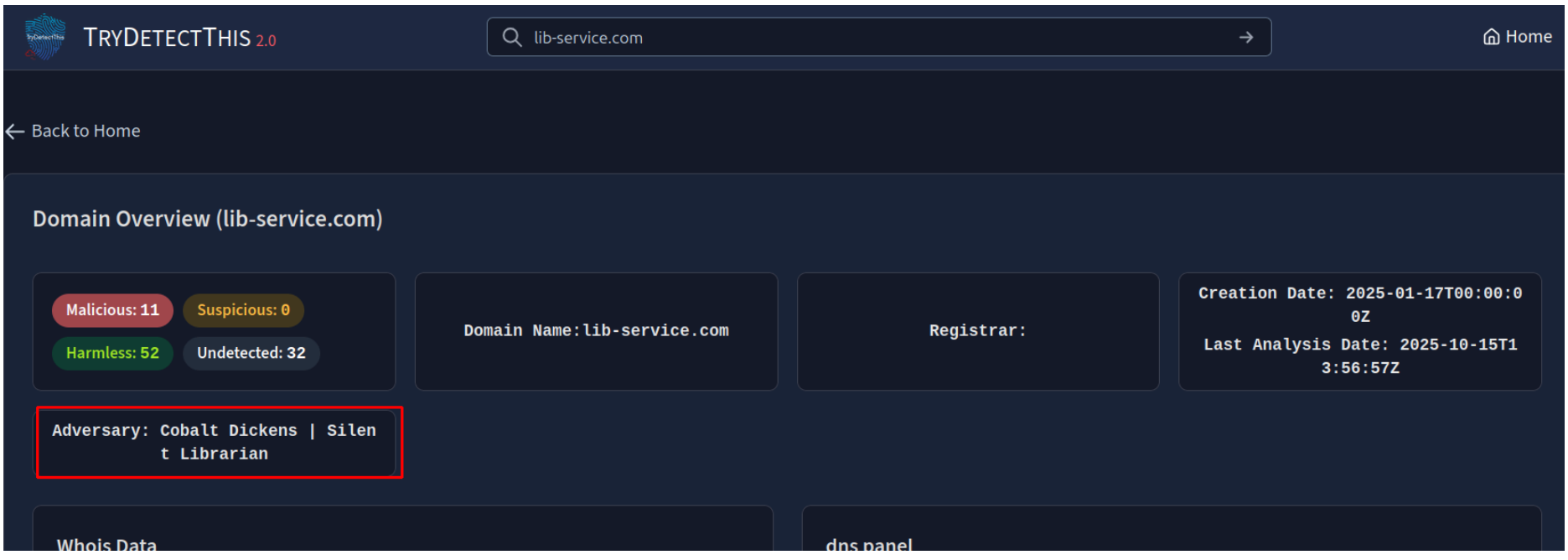
To investigate the redirect chain, I open the deobfuscated URL from Question 8 and inspect the browser's network logs (F12 → Network)



Answer: http://xn--librarytlu-13cwe32432-kwr.com:8082

11. What is the Threat Actor associated with this malicious file and/or URL?

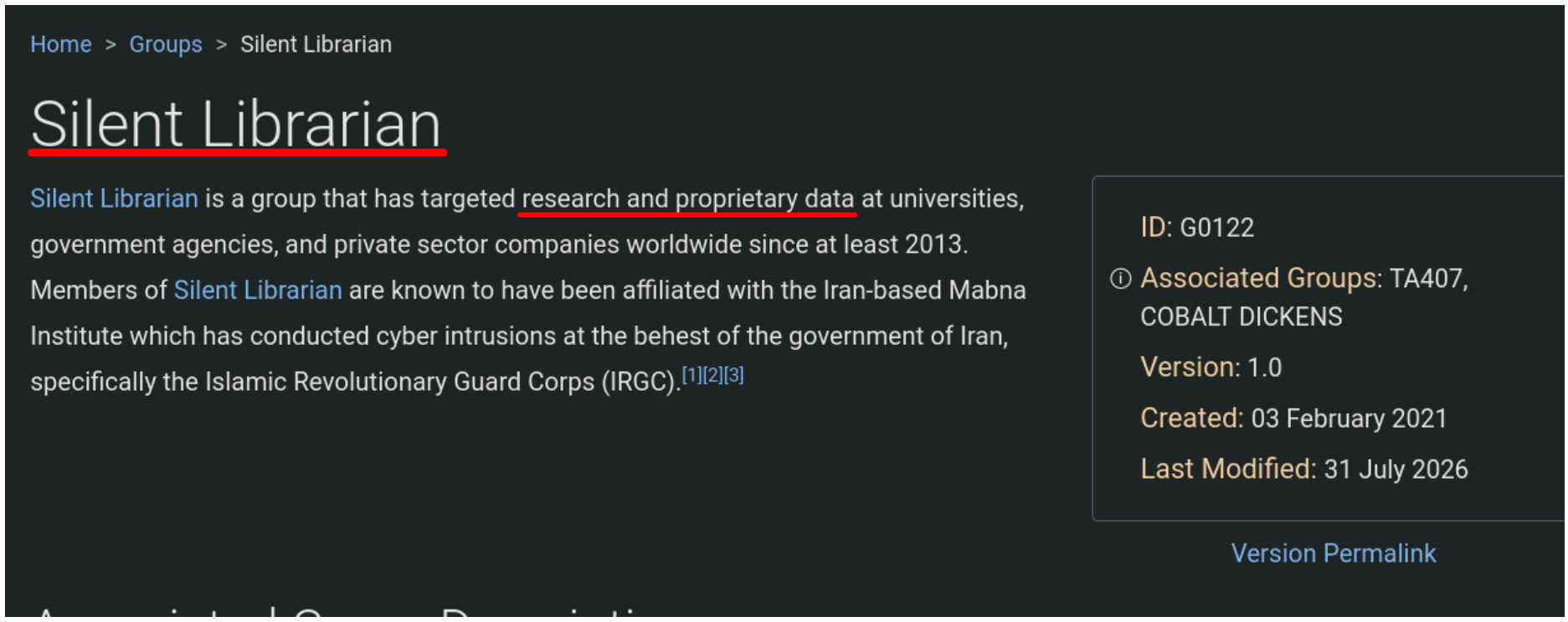
To answer this question, we access the threat intelligence database provided for this room and check the final domain



Answer: Cobalt Dickens | Silent Librarian

12. What is the main target of this Threat Actor according to MITRE?

We go to the MITRE ATT&CK website and search for the group identified in Question 11



Answer: Research and Proprietary Data